

Literature Status: BFGM Superreflexive Fixed-Point Conjecture

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Status

This is a lightweight literature-status packet. It records that Conjecture 1.6 from Bader–Furman–Gelander–Monod, *Property (T) and rigidity for actions on Banach spaces*, arXiv:math/0506361, is explicitly answered by the later paper de Laat–de la Salle, *Actions of higher rank groups on uniformly convex Banach spaces*, arXiv:2303.01405.

Source Conjecture

The source paper asks whether the higher rank groups considered in its Theorem B, and their lattices, have property $\overline{F}_B$, and hence $\overline{T}_B$, for every superreflexive Banach space B . In the parsed source this is Conjecture `C:higherrankFB`, at `source.tex:477–481`.

Supporting Answer

The supporting paper identifies this exact source. Its Theorem A is labeled as Conjecture 1.6 of the BFGM paper and states that higher rank groups and their lattices have property F_E for every super-reflexive Banach space E (`source.tex:80–82` in the parsed supporting source).

Section 7 proves the theorem. The proof reduces the superreflexive case to uniformly convex spaces using BFGM Proposition 2.13, applies the paper’s Archimedean fixed-point theorem for real rank at least two simple Lie groups, uses Lafforgue–Liao strong Banach property (T) for non-Archimedean factors, and then passes to lattices. The supporting paper also notes at `source.tex:1268` that the original barred $\overline{F}_E$ formulation and Theorem A’s F_E formulation are equivalent by BFGM Proposition 2.13.

Scope

The literature answer is affirmative for the source conjecture as stated: higher rank groups and their lattices have the fixed-point property for every superreflexive Banach space in the BFGM sense. This packet does not claim a new proof; it records an exact later-paper resolution and prevents future agents from spending solve time on a conjecture already settled in print.

References

- U. Bader, A. Furman, T. Gelander, and N. Monod, *Property (T) and rigidity for actions on Banach spaces*, arXiv:math/0506361.

- T. de Laat and M. de la Salle, *Actions of higher rank groups on uniformly convex Banach spaces*, arXiv:2303.01405.